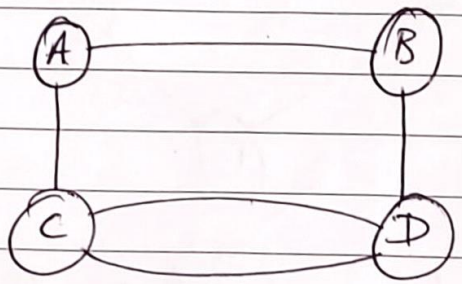
Jan 2020

Q4 Graphs + Trees.

a) see Notes for definitions...

b)

	A	B	C	D
A	0	1	1	0
B	1	0	0	1
C	1	0	0	2
D	0	1	2	0



graph G.

c) $\Sigma deg(v) = 2|E|$

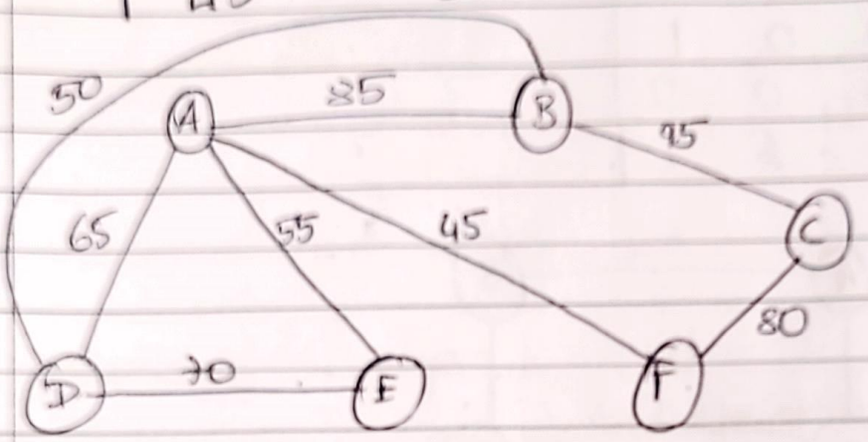
$deg(A) = 2$
$deg(B) = 2$
$deg(C) = 3$
$deg(D) = 3$
10

$E = 5$
$2|E| = 2|5| = 10$

$\therefore \Sigma deg(v) = 2|E|$
$10 = 10.$

d)

	A	B	C	D	E
B	85				
C		95			
D	65	50			
E	65			70	
F	45		80		



Prim's Algorithm to construct the MST

$T = \{A, af, F, ae, E, ad, D, bd, B, cf, E\}$

$45 + 55 + 65 + 50 + 80 = 295$

e) See notes for definitions & examples of each Graph traversal.